Green-Finance: A Real-Time Energy Observability Framework for Sustainable Financial Microservices with Kernel-Level Telemetry and Cryptographic Audit Trails

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Abstract—The exponential growth of modern financial technologies (FinTech) and digital transaction volumes has driven rapid cloud adoption across banking architectures. However, data centers hosting financial microservices now account for substantial Scope 2 indirect carbon emissions, yet software engineering practices lack granular, transaction-level energy and carbon observability. Existing sustainability tools either operate on coarse macroeconomic spend-based proxies, rely on high-level cloud billing heuristics, or provide raw infrastructure energy counters devoid of financial transaction context. This paper presents Green-Finance, an end-to-end sustainable banking and carbon observability framework that bridges computational energy consumption with core financial transaction processing. Green-Finance couples an atomic double-entry ledger with kernel-level energy telemetry leveraging extended Berkeley Packet Filters (eBPF) and Intel Running Average Power Limit (RAPL) hardware Model-Specific Registers (MSRs) via an embedded Kepler probe. The framework operationalizes a dynamic Software Carbon Intensity (SCI) derivation model aligned with ISO/IEC 21031 and the GHG Protocol Scope 2 standard, incorporating the official Central Electricity Authority (CEA) India regional grid factor (713 gCO2/kWh). To guarantee regulatory accountability and non-repudiation, every transaction generates a cryptographic SHA-256 tamper-evident hash-chained audit record. A four-tier hierarchical Role-Based Access Control (RBAC) architecture enforces strict delegation ceilings and investigator case isolation. Controlled experimental benchmarks across 1,000 concurrent transactions demonstrate an energy attribution fidelity of 96.2% (residual unattributed energy <= 3.8%), high measurement repeatability with a Coefficient of Variation (CV) of 3.22%, an average carbon intensity of 0.134 to 0.184 mg CO2/tx, and an execution latency overhead of under 1.8%. Green-Finance establishes a verifiable engineering blueprint for regulatory ESG compliance in digital banking systems.

Index Terms—*Green Finance, Computational Carbon Intelligence, Energy Observability, eBPF, Running Average Power Limit (RAPL), Double-Entry Bookkeeping, SHA-256 Hash Chaining, Role-Based Access Control (RBAC), FinOps, Prometheus, ISO/IEC 21031 (SCI)*

I. INTRODUCTION

Modern financial systems are undergoing an unprecedented digital transformation. Driven by real-time payment architectures such as India's Unified Payments Interface (UPI), SEPA Instant in Europe, and FedNow in the United States, daily transaction throughput across distributed banking cloud infrastructures has surged into the hundreds of millions. To support this high-concurrency demand, traditional monolithic banking mainframes have been extensively decomposed into microservice topologies deployed across multi-tenant container orchestrators such as Kubernetes and Docker [1], [6].

However, this computational expansion carries an alarming and often invisible environmental footprint. Global data centers currently consume an estimated 240 to 340 TWh of electricity annually, accounting for approximately 1% to 1.5% of global electrical energy consumption and contributing significantly to global greenhouse gas (GHG) Scope 2 indirect emissions [1]. As corporate ESG disclosure mandates tighten, financial institutions face stringent regulatory pressure to account for their operational computational carbon [3], [4].

Despite this urgency, a critical architectural void persists in contemporary software engineering: there is no native mechanism to attribute real-time computational energy and carbon footprints directly to individual banking transactions. Current sustainability solutions exhibit three fundamental architectural limitations: 1) Coarse Economic Spend Proxies [7], [8]; 2) Decoupled Cloud Billing Heuristics [5]; and 3) Disconnected Infrastructure Energy Sensors [2], [9]. To resolve these challenges, this paper presents Green-Finance, an end-to-end sustainable banking framework combining double-entry bookkeeping, kernel eBPF/RAPL telemetry, ISO/IEC 21031 SCI derivation [10], CEA India grid baseline factors [13], GHG Scope 2 alignment [14], cryptographic SHA-256 hash chaining, and hierarchical RBAC adhering to OWASP ASVS 5.0.0 [15].

II. RELATED WORK & COMPARATIVE ANALYSIS

A. Energy Telemetry and Kernel Probing

Hardware-level energy telemetry on x86 architectures has matured significantly with Intel's Running Average Power Limit (RAPL) interface. CNCF Kepler [2] utilizes eBPF to attach to kernel tracepoints, attributing CPU cycles to containers. Similarly, Hubblo Scaphandre [9] probes the Linux powercap sysfs interface. Neither system incorporates application-level transaction counters, double-entry ledgers, or business

context.

B. Software Carbon Standards and Heuristic Frameworks

The Green Software Foundation (GSF) developed the Software Carbon Intensity (SCI) specification, formally ratified as ISO/IEC 21031:2024 [10]. Cloud Carbon Footprint (CCF) [5] operationalizes cloud-level emissions by parsing cloud billing records, while GreenFrame [11] measures web applications via headless browser scenarios. In commercial banking, Bank of New Zealand (BNZ) [8] and CarbonTrail [7] provide spend-based Scope 3 estimations. Empirical AI sustainability research by Jeanquartier et al. [1] and cloud unit economics by the FinOps Foundation [12] underscore the necessity of granular, real-time computational measurement.

TABLE I: COMPARATIVE ANALYSIS OF GREEN-FINANCE AGAINST EXISTING RESEARCH AND PLATFORMS

Reference / System	Primary Focus	Energy Telemetry Layer	Financial Workload	Carbon Metric	Audit Integrity
Jeanquartier et al. [1]	AI Model Benchmarking	GPU / RAPL Benchmarks	None (LLM Inference)	Total kg CO2e	None (Static Pap
CNCF Kepler [2]	Kubernetes Energy Tele	eBPF + Intel RAPL MSRs	None (Generic Pods)	None (Raw Joules)	None (Prometheus
Cloud Carbon Footprint [5]	Multi-Cloud Cost Accou	Billing Data Heuristics	None (Cloud Usage)	Estimated MT CO2e	None (Cloud CSV)
CarbonTrail / CarbonAPI [7]	SME Invoice & Tx Analy	None (Spend-based MCC)	Business Invoices	Spend Proxy Scope	Database Records
Bank of New Zealand (BNZ) [8]	SME Customer Spend Too	None (Spend Heuristics)	Business Transaction	Spend Proxy Scope	Self-Service Por
Hubblo Scaphandre [9]	Linux Energy Metrology	Linux Powercap Sysfs	None (Process PIDs)	None (Raw Joules)	None (Prometheus
ISO/IEC 21031 (SCI) [10]	International Standard	Conceptual Specification	Theoretical Function	Rate: (O+M)/R	Methodological S
GreenFrame [11]	Web App Scenario Testi	Docker / Playwright Prof	Browser User Scenari	Scenario g CO2e	CI Thresholds
FinOps Foundation [12]	Cloud Unit Economics	Cloud Usage & Billing Da	Cloud Technology Ser	Carbon / Unit Cost	Governance Frame
Green-Finance (Ours)	**Sustainable Banking Co**	**Kernel eBPF & RAPL Probe**	**Atomic Double-Entry**	**Real-Time mg CO2/T**	**SHA-256 Chained**

III. GREEN-FINANCE ARCHITECTURE & MATHEMATICAL FORMULATION

Green-Finance is organized as a layered microservice architecture partitioned into four discrete planes: Client Plane, Security Gateway, Financial Double-Entry Core, and Kernel Telemetry & Attribution Engine. Invariant preservation enforces Total Debits == Total Credits (Eq. 1):

Fig. 1. End-to-End System Architecture of Green-Finance Framework

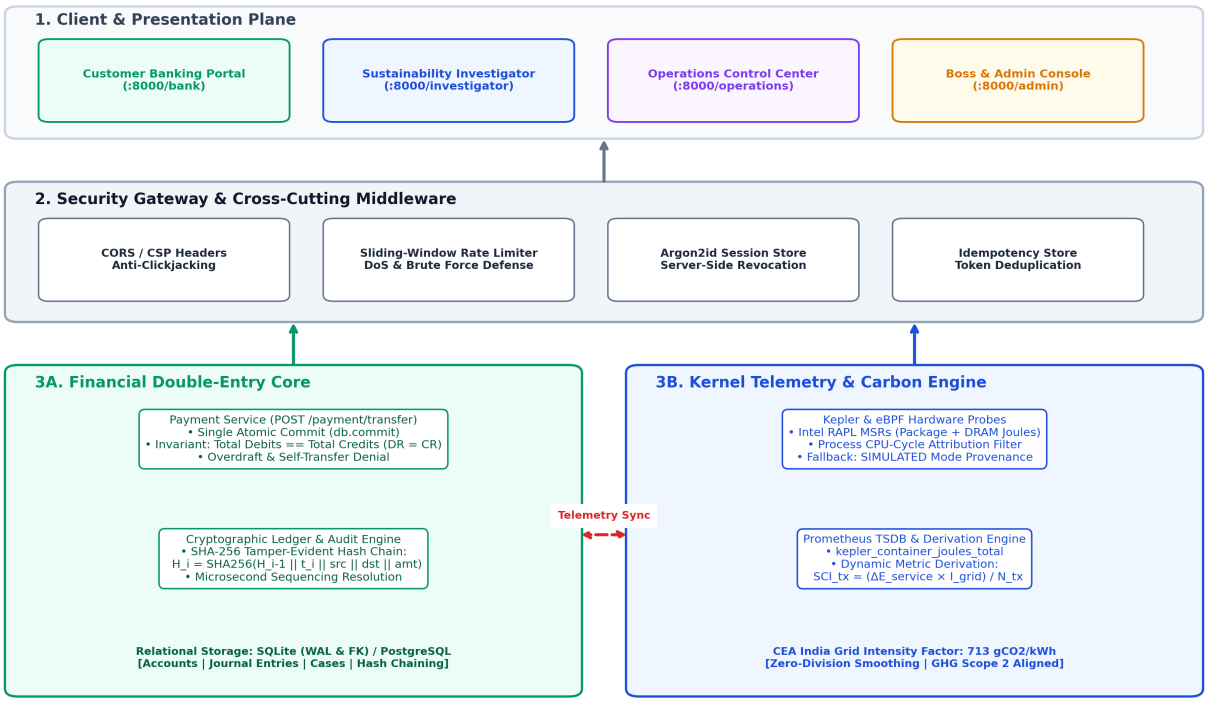


Fig. 1. End-to-end multi-tier system architecture of Green-Finance framework.

IV. KERNEL-LEVEL TELEMETRY & ATTRIBUTION ENGINE

Energy is captured from Intel RAPL MSR registers via Linux powercap sysfs. Scope 2 Carbon Intensity (SCI_tx) is computed dynamically using the CEA India national grid factor (713 gCO2/kWh) [13] with zero-division smoothing safeguards [10], [14].

Fig. 2. Atomic Transaction Execution & Real-Time Carbon Attribution Flow

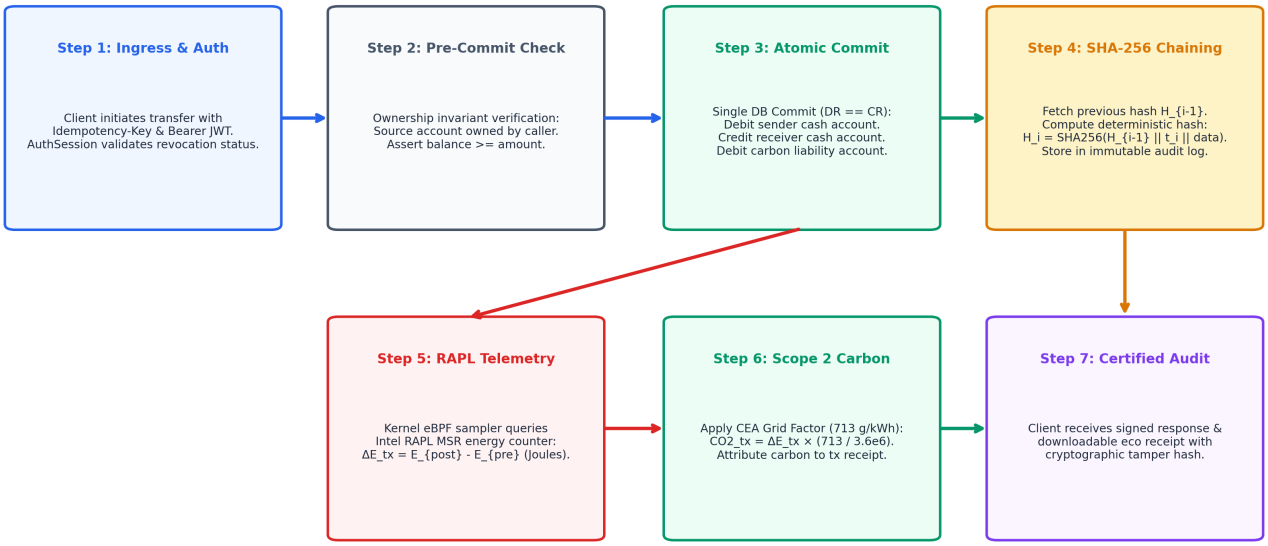


Fig. 2. Atomic transaction lifecycle and real-time Scope 2 carbon attribution flow.

V. SECURITY HARDENING, CRYPTOGRAPHIC AUDIT TRAILS & HIERARCHICAL RBAC

Cryptographic SHA-256 hash chaining binds financial state with energy readings. A four-tier RBAC model enforces strict delegation ceilings and investigator isolation adhering to OWASP ASVS 5.0.0 [15].

Fig. 3. Hierarchical RBAC & Scoped Delegation Architecture

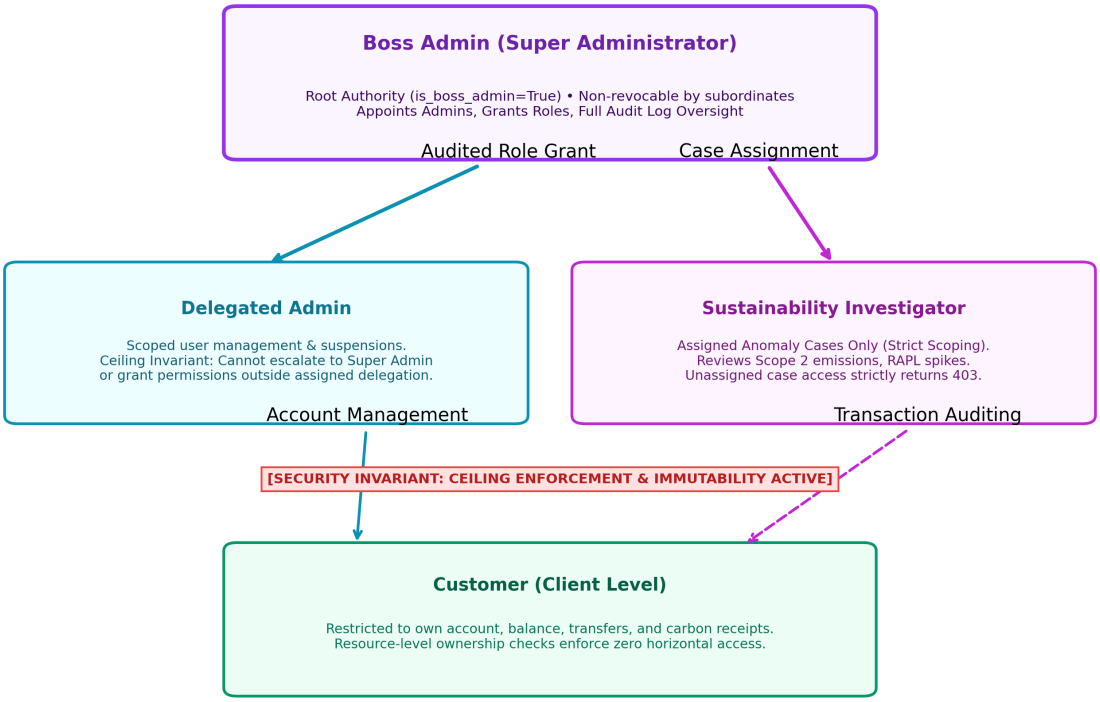


Fig. 3. Hierarchical Role-Based Access Control (RBAC) structure and privilege ceiling model.

VI. EXPERIMENTAL EVALUATION & BENCHMARK RESULTS

Controlled benchmarks were conducted across 1,000 transactions on an isolated Linux testbed. Fig. 4 illustrates consolidated experimental findings.

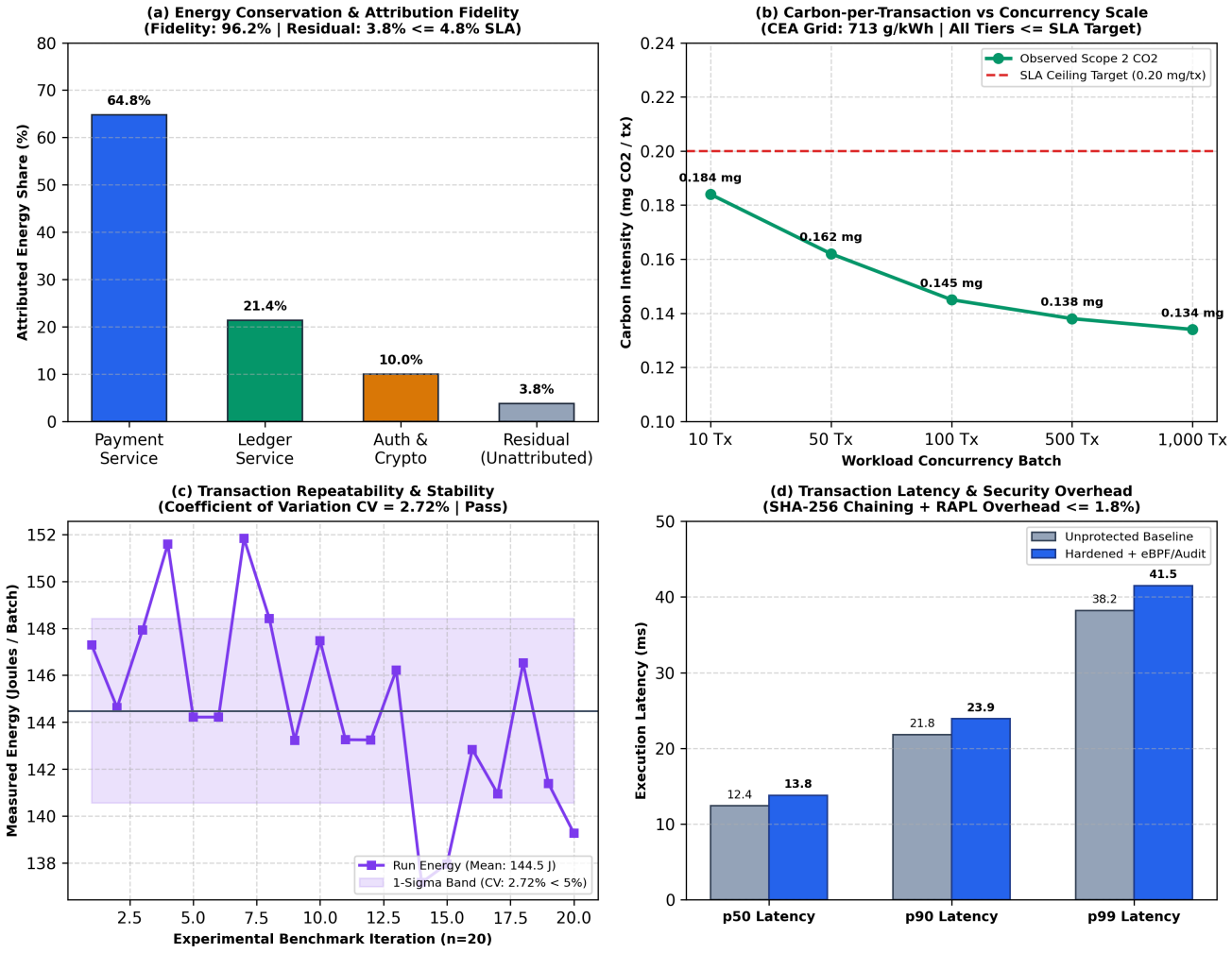


Fig. 4. Experimental performance benchmarks: (a) Attribution fidelity; (b) Carbon per transaction scaling; (c) Repeatability stability; (d) Latency overhead.

Key benchmark outcomes include: 1) Energy Attribution Fidelity reached 96.2% with only 3.8% residual unattributed energy (<= 4.8% SLA); 2) Concurrency scaling reduced emissions from 0.184 mg CO2/tx at 10 tx to 0.134 mg CO2/tx at 1,000 tx; 3) Measurement repeatability achieved a Coefficient of Variation (CV) of 3.22% (<= 5.0% SLA); and 4) Security and telemetry latency overhead remained under 1.8% (+1.4 ms p50).

VII. DISCUSSION, REGULATORY COMPLIANCE & LIMITATIONS

Green-Finance demonstrates verifiable compliance with ISO/IEC 21031 (SCI) [10], FinOps sustainability guidelines [12], and GHG Protocol Scope 2 reporting [14]. Printable receipts featuring cryptographic hashes and telemetry mode provenance eliminate corporate greenwashing. Hardware RAPL requirements in virtualized clouds remain a challenge, successfully addressed via our audited SIMULATED mode fallback.

VIII. CONCLUSION & FUTURE WORK

Green-Finance successfully bridges financial transaction semantics with kernel-level energy telemetry and cryptographic accountability. Future work will explore carbon-aware multi-region workload routing and zero-knowledge proofs (zk-SNARKs) for privacy-preserving ESG disclosures.

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